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Antibiotic resistance: a global crisis, problems and solutions

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ABSTRACT

Healthy state is priority in today's world which can be achieved using effective medicines. But due to overuse and misuse of antibiotics, a menace of resistance has increased in pathogenic microbes. World Health Organization (WHO) has announced ESKAPE pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* spp.) as the top priority pathogens as these have developed resistance against certain antibiotics. To combat such a global issue, it is utmost important to identify novel therapeutic strategies/agents as an alternate to such antibiotics. To name certain antibiotic adjuvants including: inhibitors of beta-lactamase, efflux pumps and permeabilizers for outer membrane can potentially solve the antibiotic resistance problems. In this regard, inhibitors of lytic domain of lytic transglycosylases provide a novel way to not only act as an alternate to antibiotics but also capable of restoring the efficiency of previously resistant antibiotics. Further, use of bacteriophages is another promising strategy to deal with antibiotic resistant pathogens. Taking in consideration the alternatives of antibiotics, a green synthesis nanoparticle-based therapy exemplifies a good option to combat microbial resistance. As horizontal gene transfer (HGT) in bacteria facilitates the evolution of new resistance strains, therefore identifying the mechanism of resistance and development of inhibitors against it can be a novel approach to combat such problems. In our perspective, host-directed therapy (HDT) represents another promising strategy in combating antimicrobial resistance (AMR). This approach involves targeting specific factors within host cells that pathogens rely on for their survival, either through replication or persistence. As many new drugs are under clinical trials it is advisable that more clinical data and antimicrobial stewardship programs should be conducted to fully assess the clinical efficacy and safety of new therapeutic agents.

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Introduction

Antibiotic resistance (ABR) is a worldwide problem leading to many health challenges, as a consequence of processes involving the horizontal transfer of bacterial gene or the bacteria between animal-human, human-human or with the environment. Bacteria could acquire the resistance *via* uptake of foreign DNA or *via* mutation in the precursory genome (Larsson and Flach 2022). Antibiotics have historically served as a treatment option for a wide range of life-threatening infections. However, due to overuse and misuse of antibiotics or because of various evolutionary laws, ABR is being developed in less economically developed as well as in economically developed

countries. In addition, the lack of rapid diagnostic methods to identify resistant bacterial pathogens and AMR genes in clinical settings has resulted in the often-unnecessary use of broad-spectrum antibiotics (Akova 2016; De Oliveira et al. 2020). In this regard, WHO has described a list of 'ESKAPE' organisms which includes those microbes which are ABR organisms and are frequently involved in causing nosocomial opportunistic infections. Patients who are either severely ill or immunocompromised have greater chance to acquire healthcare-associated infections *via* ESKAPE pathogens (Akova 2016). Antibiotics have played a pivotal role in human healthcare system by reducing illness and death rates. For instance, in case

of Tuberculosis, which is a contagious infection commonly causing more than 10 lakh cases per year was greatly edged off *via* the use of streptomycin and kanamycin (Ogawara 2019). However, due to continuous resistance development in bacteria, once again the situation got reverted back and is getting worsen each passing day (Zhen et al. 2019). Various mechanisms underlie the development of multidrug resistance (MDR) including efflux transporters, modification of target, remodeling of membrane, modification or destruction of antibiotic inside bacterial cell. Such mechanisms are either acquired through mutation, HGT or through adaptive antibiotic resistance (Blair et al. 2015; Munita and Arias 2016). Furthermore, there have been reports related to highly transmissible and drug-resistant ringworm infections in New York, USA (NBC news 2023). These cases were initially discovered in a 47-year-old woman who acquired a severe ringworm infection, known as tinea, during her visit to Bangladesh. It was determined that the woman's infection was caused by a relatively new species of ringworm-causing fungus called *Trichophyton indotineae*. The rapid spread of infections caused by this drug-resistant fungus has been observed in South Asia over the past decade, likely influenced by the excessive use of medications such as topical antifungals and corticosteroids for treatment. Similar to drug-resistant bacteria, drug-resistant fungi pose a significant public health threat. Notably, instances of *Candida auris* (another type of drug-resistant fungus which causes infection), have been increasingly reported in healthcare facilities across the United

States (Cristina et al. 2023). This particular infection is extremely challenging to treat and can have fatal outcomes.

Antibiotics have served an important component in treatment of various fatal, virulent and devastating bacterial infections but because of emergence of ABR there is again a threat on the beneficial effects of antibiotics which it achieved in last several years (Ventola 2015). Therefore, we have initiated to apprise with this global issue and plausible means to deal with it. We present the AMR mechanism and the urgency to cope up with them using possible strategies and alternatives of antibiotics against which human pathogens have developed resistance.

Problems of antibiotics resistance

Antimicrobial resistance a global concern

Four main mechanisms which gave bacteria the power to gain resistance against several drugs have been described. These include remodeling of antibiotics which involved either destruction or modification, remodeling of target site, efflux of drug component, and remodeling of membrane decreasing its permeability for antibiotics (Figure 1) (Laws et al. 2019). It's not very necessary that a single bacterium has access to a single mechanism, it may develop different combinations of such mechanisms which leads to even dangerous circumstances as this gives the microbe the ability to resist against more than one class of antibiotics, gaining higher

Figure 1. General antibiotic mechanism developed by bacteria. **(1)** Antibiotic efflux; **(2)** decreased membrane permeability; **(3)** antibiotic modification; **(4)** modification of target site.

resistance (Nikaido 2009). Bacteria can acquire resistance either through HGT which involves bacterial genetic recombination. In this kind of evolution, susceptible bacteria achieve the resistance gene from other bacteria which has already developed the resistance. In this regard, HGT helps microbes to exchange their genetic material containing antibiotic-resistant genes (ARGs) between various species (Le Roux and Blokesch 2018). For instance, superbugs developed resistance against all approved antibiotics used to treat infections, primarily due to transfer of ARG on their plasmid through HGT (Malhotra-Kumar et al. 2016). In the case of vertical evolution, mutation occurs in the genome of bacteria which allows dominant, resistant pathogen to be selected in a given population of other bacteria (Laws et al. 2019). Microorganisms also assemble together to form a unique structure known as biofilm which helps them to survive in the harsh environment of antimicrobials (Sharma et al. 2020; Verma et al. 2023). There are various factors which control the formation of biofilm including nutrient availability, quorum sensing, cell-to-cell communication. It is important to note that biofilm formation not only poses a threat to human and animal health but also to food industry (Dai et al. 2021; Rather et al. 2021).

According to the Center for Disease Control (CDC), USA, every year almost 23,000 deaths occur in the USA owing to AMR. It has been predicted that the death number would rise to millions in the upcoming decades (Durand et al. 2019). Bacterial AMR leads to roughly 1.27 million deaths every year. Additionally, there are roughly 4.95 million deaths annually linked to bacterial AMR, which is more than the total deaths caused by HIV, tuberculosis (TB), and malaria combined. It is a matter of great concern that one out of every five deaths caused by AMR happens in children under five years old, often because of infections that used to be curable (WHO 2023a). According to the WHO, if no measures are taken, it is projected that the number of deaths could rise to 10 million by the year 2050 (WHO 2019a). Researchers have devised numerous strategies to combat the resistant bacteria, including the discovery of new antimicrobial agents, finding unique antibiotic targets which show no resistance, modifying the structure of preexisting antibiotics. Researchers have also discovered some latest chemical structures which have new mechanisms of action as an antibiotic to tackle the problem of resistance (Darby et al. 2023). This growing concern of ABR demands an urgent strategy to overcome the issue of resistance, many new treatments have also been succeeded against Gram-negative bacteria including β -lactamase inhibitor antibiotic adjuvants;

metal-based antibacterial agents; few plants derived agents such as Odilorhabdins, DCAP; quorum sensing (QS) inhibitors etc (Breijyeh et al. 2020).

Impact of AMR on global economy

AMR on one hand effects patient's health, while on other hand it causes economic burden. Significant effects have also been observed on nation's economic condition (Lushniak 2014). Azithromycin, one of the antibiotics which was overused during pandemic situation of COVID-19. In this regard, WHO has predicted that ABR can be developed by bacteria in upcoming years as this medicine was taken by patients who were symptomatic as well asymptomatic without even recommendation of health authorities. During the ongoing COVID-19 pandemic, despite the low rates of secondary infections related to bacteria and fungus, high percentage of patients were treated with broad-spectrum antibiotics which indicated the high consumption of antibiotics. Due to the paucity of data, it was suggested that detail clinical studies are required focusing the antimicrobial therapy (WHO 2019b; Rawson et al. 2020). Due to over-prescription of antibiotics during COVID-19, WHO prohibited the unindicted use of medicine in patients with mild symptoms (Hsu 2020). Recently, World Bank on global economy (2017–2050) based on current AMR patterns has predicted that the countries having low GDP (Gross Domestic Product) will be unsympathetically affected if AMR continues to rise. It is estimated that the world GDP will drop to 3.8% by the year 2050 (World Bank 2023). Also, if AMR problem persists as now, then a 1.1% fall in the world's GDP with \$1 trillion annual GDP shortfall after 2030 can be seen. Ahmad and Khan (2019) suggested that AMR will have an impact on various factors by the year 2050 that also includes GDP. They have further suggested that there will be a fall by 1.1% and 3.8% in the countries with low and high AMR problems respectively. This indicates impact on poverty which would increase extremely in low as well as in high impact AMR countries. Of the additional 28.3 million people falling into extreme poverty in the high impact AMR scenario, the vast majority (26.2 million) would live in low-income countries; impact on world trade would also be affected by 2050 as the trade will narrow down by 3.8% in high AMR countries and by 1.1% in low AMR countries. A global increase in healthcare costs will also be seen which would range from US \$300 billion to more than US \$1 trillion per year (Ahmad and Khan 2019).

Environment as a resource as well as manager of ABR

The trouble during the course of evolution of resistance is that external environment serves as a source a pool of genomes responsible for developing antibacterial resistance. Evolution of resistance against any antibiotic was found to be either by foreign DNA entry through HGT or by mutation in bacteria. However, there is less contribution of external environments related to mutation-based evolution of resistance for most pathogens. Several microorganisms resided in the external environment contributes to the transfer of ARGs by mediating association with insertion sequences or relocating the gene to plasmid, transposons, and HGT within the environment. In addition, it has been suggested that domestic animals and human microbiota also become the source of ARGs transmission between themselves and with the environment (Larsson and Flach 2022). Environment components like water, air, food etc contaminated with the fecal matter provide the gene exchange source between environmental microbes and intestinal tract resident human and animal bacteria (Pal et al. 2016; Leonard et al. 2018; Zhu et al. 2019). It has been suggested that AMR is, in part, happened due to horizontal transfer of antibiotic resistance genes (ARGs), typically mediated by plasmids. Many of the plasmid-mediated resistance genes in pathogens originate from environmental, animal or human habitats. However, these factors yet to be identified, collectively determine the emergence and transfer of plasmid-mediated AMR within and between habitats at the local and global scale (CastañedaBarba et al. 2023).

The natural environment is polluted through improper disposal of unused antibiotics, wrong handling of expired medicine (Anwar et al. 2020). The detection of macrolides, sulfonamides, ciprofloxacin, tetracycline etc in aquatic environment results in increasing case of resistance among various bacteria (Wang et al. 2020). Also, external environment serves as a manager as it can provide ecological niche for fungus and bacteria that contributes a lot in advancing drug development by giving various novel candidate antibiotic molecules (Lewis 2013). Therefore, environment provides better understanding of evolution of resistance at temporal and regional situation. Therefore, various important factors that spread the ABR and drive the evolution of environment needs identification which ultimately manage the overall crisis.

ABR in developed and developing countries

Whole world is facing the problem of ABR which is rising day by day to a great extent. Because of emergence

of new resistance mechanisms common infectious diseases are also becoming untreatable, posing a threat. In countries without proper treatment guidelines and less developed health infrastructure like in South East Asian, Sub-Sahara and African countries, where people consume antibiotics without any prescription, the problem of ABR is exacerbating. Many a times in such countries either health workers over-prescribes the antibiotics or it is available for over-the-counter consumption. Hospitals and community play a very big role in the spread of resistant strains. AMR is a growing pandemic and a potential threat to public health in developed as well as in developing countries. According to the Global Research on Antimicrobial Resistance (GRAM) report, based on estimates from 204 countries and territories, (Tropical Medicine 2023) it is mentioned that in 2019, approximately 4.95 million deaths were potentially linked to drug-resistant infections, with at least 1.27 million directly attributed to bacterial infections resulting from ABR. India specifically exhibits high rates of ABR among which bacteria are commonly associated with infections in both community and healthcare settings. Worldwide distribution of reservoir of New Delhi Metallo-beta-lactamase (NDM) variants is about 58.15% in Asian countries like China, India, Sri Lanka and Bangladesh. In European countries it is 16.8%, in USA and Africa it is 10.8% each, while in Australia it is 1.6% (Farhat and Khan 2020). Almost all countries are facing ABR problems whether it's high income or a low-income country. Occurrence of ABR was reported more in case of third world countries as compared to developed countries. A diverse ABR spectrum observed in developed and developing countries is primarily due to differential strategies adopted and implemented for antibiotic consumption (Mahmudul 2021). Usually, prescription of antibiotics is logical and targeted to specific bacteria in developed countries whereas it is unselective to the causation of disease in developing countries. In 2015, the WHO initiated the Global Antimicrobial Resistance and Use Surveillance System (GLASS) as a collaborative effort (Antimicrobial resistance through the looking-GLASS 2023). The primary objective of this program was to assess the global burden of AMR through optimized surveillance. However, this report highlights the limited testing coverage in low-income and middle-income countries (LMICs) and emphasizes the need for more data collection. The rates of AMR for many pathogen-drug combinations remained stable in 2020 as compared to previous years. However, concerning trends were observed, as more than 15% increase in AMR rates for azithromycin in case of *Neisseria gonorrhoeae*, as well as resistance against meropenem and third-generation cephalosporins in case of *Escherichia*

coli, and ciprofloxacin resistance in case of *Salmonella* spp. The GLASS program emphasizes the necessity for further research on AMR surveillance. Persistent high levels of carbapenem resistance reported in *Klebsiella pneumoniae* and *Acinetobacter* spp. causing bloodstream infections are alarming indicators of increasing treatment failures associated with carbapenem-resistant infections due to the rise of MDR. Notably, high levels of AMR in bloodstream infections were reported irrespective of testing coverage (Antimicrobial resistance through the looking-GLASS 2023). These findings underscore the worrisome trend of common infections becoming increasingly resistant to the treatment, as evidenced in urinary tract infections and *N. gonorrhoeae* infections. Table 1 summarizes ABR bacteria and antibiotics used in developed and developing countries.

Solutions of the antibiotic's resistance

Epidemiology/surveillance of AMR

Before the development or search of any therapeutics we should know the type of genus, species and strains circulating or their prevalence and type of infections caused by them in that particular area or country. For instance, in the case of *Klebsiella* genus of *Enterobacteriaceae* family which causes hospital and community infections, different global clinical pattern has been observed (Podschun and Ullmann 1998; Ko et al. 2002). The *Klebsiella* genus can colonize intestinal tract and other mucosal surfaces of humans hence causes urinary tract infections, pneumonia, bloodstream infections, and sepsis (Selden et al. 1971; Podschun and Ullmann 1998; Afroza 2006; Meatherall et al. 2009; Huynh et al. 2020). Infections occur most commonly in the senior adults, neonates and in case of immunocompromised patients. Amongst, *Klebsiella* genus, *K. pneumoniae*, *K. quasipneumoniae*, *K. variicola*, and *K. oxytoca* are the most notable pathogens (Pendleton et al. 2013; Singh et al. 2016; Mathers et al. 2019; Rodríguez-Medina et al. 2019). It is important to note that hypervirulent *K. pneumoniae* causes life threatening infections due to the production of siderophores and the capsule expression regulator hypermucoviscous phenotype (RmpA) (Shon et al. 2013). In recent years, infections caused by both multidrug resistance and hypervirulence have been reported which leads to a concern over a potential rise of serious untreatable infections (Gu et al. 2018; Shen et al. 2019). In this scenario, pathogen surveillance can provide a powerful tool to understand the evolution and spread of resistant bacteria and defining their clinically relevant features (David et al. 2019). With the advent of

genomics, computational biology and bioinformatics, there is a growing need to use whole-genome sequencing (WGS), to identify antibiotics resistant species in hospitals and community (Argimón et al. 2021; Sánchez-Busó et al. 2021). Recently pathogen watch was used to analyze *Klebsiella* genomes from low- and middle-income countries which participated in a program conducted by the National Institute for Health Research, Global Health Research Unit (GHRU) on genomic surveillance of AMR (Argimón et al. 2021).

Amongst ESKAPE organisms *A. baumannii* has been shown to develop resistance against a broad range of antimicrobials, including carbapenems and have been found responsible for epidemics worldwide (Wong et al. 2017; Gaiarsa et al. 2019; Hamidian and Nigro 2019). Previously, the identification of *Acinetobacter* isolates at the species level has been found to be difficult by using classical phenotypic methods (Bouvet and Grimont 1986; Cools et al. 2019). In recent years, Matrix-Assisted Laser Desorption-Ionization-Time of Flight (MALDI-TOF) mass spectrometry or genotypic methods, which use partial beta-subunit of RNA polymerase (*rpoB*) sequencing or ribosomal MLST, Pasteur MLST scheme and phylogenomic analysis have been performed to identify AMR genes in many species of *Acinetobacter* genus (Gundi et al. 2009; Jolley et al. 2012; MarY-Almirall et al. 2017; Šedo et al. 2018; Migliaccio et al. 2023). In our opinion, in case of all ESKAPE organisms, similar techniques as mentioned in previous sections will be useful in order to identify species, strains and AMR genes in developed and economically less developed countries.

Alternate to antibiotics

Antibiotic adjuvants are the chemicals that have either very little or have no antibiotic like activity but, when coupled with some drug compound as the part of a synergistic medication treatment, they boost the compound's efficacy or inhibit the bacteria's resistance. Antibiotic adjuvants have resurrected the use of various antibiotics against resistant microbes or infections caused by the resistant pathogens, reducing novel medicines with new targets, which is a costly and a daunting task (Gill et al. 2015). To block the mechanisms developed by bacteria for ABR, mainly antibiotic adjuvants are categorized as (a) β -lactamase inhibitors (BLIs) (b) inhibitors of efflux pumps, and (c) permeabilizers for outer membrane. In addition, novel possible inhibitors are being developed to target pathogenicity, or the bacterium's ability to cause infections, in order to improve the antimicrobial effect of medications. This method entails identifying proteins, genes, and other

Table 1. Summary of antibiotic resistance bacteria with the resistant antibiotic in developed and developing countries.

Country	Income level	Antibiotic resistant Bacteria	Antibiotics/class of antibiotics	Reference
Afghanistan	Low	<i>A. baumannii</i>	Carbapenems and colistin	Ahmadzai et al. 2022
		<i>P. aeruginosa</i>	Ampicillin, amoxicillin, and erythromycin, ceftriaxone, ceftazidime, cefixime, and sulfamethoxazole	Ahmadzai et al. 2022; Joya et al., 2022
		<i>S. aureus</i>	Penicillin, erythromycin, ciprofloxacin, cotrimoxazole, tobramycin, gentamicin, chloramphenicol, doxycycline, methicillin, clindamycin, rifampicin, cotrimoxazole	Naimi et al. 2021; Ahmadzai et al. 2022
Nepal		<i>K. pneumoniae</i>	Ampicillin, amoxicillin, and erythromycin, ceftriaxone, ceftazidime, cefixime, and sulfamethoxazole	Ahmadzai et al. 2022; Joya et al., 2022
		<i>E. coli</i>	Ampicillin, amoxicillin, and erythromycin, ceftriaxone, ceftazidime, cefixime, and sulfamethoxazole	Joya et al., 2022
		<i>S. aureus</i>	Clindamycin, gentamycin, tobramycin, ciprofloxacin, ofloxacin, cefuroxime, ceftriaxone, ceftazidime, chloramphenicol, cotrimoxazole	Shrestha et al. 2021; Karn et al. 2021
Nigeria		<i>K. pneumoniae</i>	Gentamycin, amikacin, tobramycin, ciprofloxacin, ofloxacin, cefuroxime, ceftriaxone, ceftazidime, chloramphenicol, cotrimoxazole	Karn et al. 2021
		<i>P. aeruginosa</i>	Gentamycin, amikacin, tobramycin, ciprofloxacin, ofloxacin, cefuroxime, ceftriaxone, ceftazidime, chloramphenicol, cotrimoxazole	Karn et al. 2021
		<i>S. enterica</i>	Cefotaxime, ampicillin	Fowler et al. 2021
		<i>Acinetobacter spp.</i>	Cephalosporin, carbapenems, colistin and polymyxins, penicillins and cephalosporins, fluoroquinolones	Mahto et al. 2021
		<i>E. coli</i>	Norfloxacin, amoxicillin–clavulanic acid, co–trimoxazole, ceftriaxone, ciprofloxacin	Kushwaha et al. 2021
		<i>Aeromonas jandaei</i>	Colistin	Komeda et al. 2022
		<i>E. coli</i>	Cefuroxime, cefotaxime, amoxicillin, imipenem, ceftriaxone, cefalexin, ampicillin, ciprofloxacin, nalidixic acid, gentamycin, nitrofurantoin, kanamycin, chloramphenicol, pefloxacin, ofloxacin, streptomycin, ceftazidime, tetracycline, amoxicillin, trimethoprim, co–trimoxazole	Mahmudul, 2021
		<i>Salmonella spp.</i>	Ampicillin, cefotaxime, chloramphenicol, trimethoprim–sulfamethoxazole, ofloxacin, ciprofloxacin, co–trimoxazole, tetracycline, efazidime, ceftriaxone	Mahmudul, 2021
		<i>Staphylococcus spp.</i>	Streptomycin, gentamycin, tetracycline, cotrimoxazole, erythromycin, cloxacillin, chloramphenicol, augmentin, imipenem, ceftriaxone, cefoxitin, ciprofloxacin, erythromycin, cefalexin co–trimoxazole, nalidixic acid, ampicillin, vancomycin, azithromycin, cefuroxime, amoxicillin, ceftazidime	Sundaresan et al. 2022
			B–lactams, fluoroquinolones, aminoglycosides	Mahmudul, 2021; Argimón et al. 2022
India	Low and middle	<i>K. pneumoniae</i>	Sulphamethazole, chloramphenicol amikacin ceftazidime, oxytetracycline, nalidixic acid, third–generation cephalosporin	Shaw et al. 2022
		<i>S. typhi</i>	Tetracycline and chloramphenicol	Yadav et al. 2022
		<i>Vibrio cholerae O1</i>	Sulfonamides, tetracycline	Kumar et al. 2022
		<i>Clostridium perfringens</i>	Metronidazole and clarithromycin	Sen et al. 2022
		<i>Helicobacter pylori</i>	Nalidixic acid, quinolones, tetracycline	Mahmudul, 2021; Brahma et al. 2022
		<i>Campylobacter jejuni</i>	Penicillin, ciprofloxacin, tetracycline, erythromycin ampicillin, tetracycline amoxicillin, erythromycin polymyxin-b, cefoxitin novobiocin, oxacillin, gentamycin, clindamycin, erythromycin, rifampicin, tetracycline, oxacillin, cefoxitin, teicoplanin.	
		<i>E. coli</i>	Amoxicillin, ampicillin, cephalixin, colistin, cefoxitin, chloramphenicol, neomycin, ciprofloxacin, co–trimoxazole, trimethoprim, amoxicillin, erythromycin, rifamycin, streptomycin, doxycycline, sulfamethoxazole, nalidixic acid, tetracycline, gentamicin	Mahmudul, 2021; Khalid et al. 2020
			Carbapenem	Tiwari et al. 2012
		<i>A. baumannii</i>	Streptomycin, trimethoprim	Manzoor et al. 2016
		<i>Lactobacillus plantarum</i>	Streptomycin, trimethoprim	Manzoor et al. 2016
Pakistan		<i>L. acidophilus AS-1</i>	Kanamycin, ampicillin, neomycin, ciprofloxacin, penicillin, chloramphenicol, erythromycin, vancomycin, and trimethoprim	Manzoor et al. 2016
		<i>E. fecalis</i>	Kanamycin, ampicillin, neomycin, ciprofloxacin, penicillin, chloramphenicol, erythromycin, vancomycin, and trimethoprim	Manzoor et al. 2016
		<i>E. coli</i>	Cefoxitin, gentamicin, oxacillin, penicillin, levofloxacin, moxifloxacin, trimethoprim, ciprofloxacin, erythromycin, and vancomycin	Manzoor et al. 2016
Bangladesh		<i>Staphylococcus spp.</i>	Colistin, nalidixic acid, cefixime, co–trimoxazole, ceftazidime, gentamicin, amikacin, imipenem, ciprofloxacin, azithromycin, cefuroxime, cefotaxime, ceftriaxone, meropenem, nitrofurantoin, levofloxacin, meropenem	Mahmudul, 2021; Manzoor et al. 2016
		<i>E. coli</i>	Colistin, nalidixic acid, cefixime, co–trimoxazole, ceftazidime, gentamicin, amikacin, imipenem, ciprofloxacin, azithromycin, cefuroxime, cefotaxime, ceftriaxone, meropenem, nitrofurantoin, levofloxacin, meropenem	Mahmudul, 2021
		<i>Salmonella spp.</i>	Ciprofloxacin, ceftriaxone, azithromycin, clindamycin	Mahmudul, 2021
		<i>Staphylococcus spp.</i>	Nalidixic acid, cefixime, meropenem, oxacillin, gentamicin, ceftazidim, tocefotitin, etracylin, cefoxitin, ciprofloxacin, chloramphenicol, clindamycin, cefotaxime, levofloxacin	Mahmudul, 2021



Country	Income level	Antibiotic resistant Bacteria	Carbapenem	Antibiotics/class of antibiotics	Reference
Brazil	Upper and middle	<i>A. baumannii</i>	Carbapenem		Barcelos et al. 2022
		<i>H. pylori</i>	Clarithromycin		Benigno et al. 2022
		<i>S. heidelberg</i>	Third generation cephalosporins, tetracyclines and/or sulfonamides		Alikhan et al. 2022
		<i>S. Minnesota</i>	Third generation cephalosporins, tetracyclines and/or sulfonamides		Alikhan et al. 2022
		<i>M. tuberculosis</i>	Ofloxacin, capreomycin, kanamycin, amikacin		Sousa et al. 2022
Malaysia		<i>K. pneumoniae</i>	Ampicillin-sulbactam, piperacillin-tazobactam, amoxicillin-clavulanate, cefazolin, cefotaxime, ertapenem, nitrofurantoin, sulfamethoxazole, tetracycline, ciprofloxacin, levofloxacin		Nakamura-Silva et al. 2022
		<i>E. coli</i>	Aminopenicillin		Naeemudeen et al. 2021
		<i>A. baumannii</i>	Amoxicillin-clavulanate, ampicillin, cefoperazone, cefuroxime		Naeemudeen et al. 2021
		<i>Klebsiella pneumoniae</i>	Ampicillin, amoxicillin-clavulanate acid, aztreonam		Naeemudeen et al. 2021; Mobasser et al. 2021
		<i>S. aureus</i>	Penicillin, gentamicin, vancomycin		Naeemudeen et al. 2021
South Africa		<i>E. faecium</i>	Erythromycin, tetracycline		Naeemudeen et al. 2021
		<i>K. pneumoniae</i>	Third-generation cephalosporin (cefazidime, ceftioxaone, cefotaxime)		Roberts et al. 2022; Malinga et al. 2022
		<i>E. coli</i>	Third-generation cephalosporin (cefazidime, ceftioxaone, cefotaxime)		Roberts et al. 2022
		<i>S. enterica</i>	Ceftazidime, ceftioxaone, cefotaxime, ciprofloxacin, ofloxacin, nalidixic acid, azithromycin		Roberts et al. 2022
		<i>M. tuberculosis</i>	Isoniazid, fluoroquinolones, ethionamide, amikacin, kanamycin, capreomycin		Penn-Nicholson et al. 2022
Australia	High	<i>A. baumannii</i>	Meropenem, imipenem		Roberts et al. 2022; Perovic et al. 2022
		<i>S. pneumoniae</i>	Penicillin, erythromycin		Roberts et al. 2022; Javaid et al. 2022
		<i>Mycoplasma genitalium</i>	Fluoroquinolone, macrolide		Mahlangu et al. 2022
		<i>Campylobacter jejuni</i>	Tetracycline, nalidixic acid, ciprofloxacin, erythromycin, chloramphenicol, ampicillin, gentamycin		Paintsil et al. 2022
		<i>C. coli</i>	Tetracycline, nalidixic acid, ciprofloxacin, erythromycin, chloramphenicol, ampicillin, gentamycin		Paintsil et al. 2022
Canada		<i>P. aeruginosa</i>	Ceftazidime		Wozniak et al. 2022
		<i>K. pneumoniae</i>	Third-generation cephalosporin		Wozniak et al. 2022
		<i>E. coli</i>	Third-generation cephalosporin		Wozniak et al. 2022
		<i>S. aureus</i>	Methicillin		Wozniak et al. 2022
		<i>E. faecium</i>	Vancomycin		Wozniak et al. 2022
Japan		<i>Neisseria gonorrhoeae</i>	Azithromycin		Sawatzky et al. 2015
		<i>Carbapenemase-producing Enterobacteriales (CPE)</i>	Imipenem/relebactam		Bartoszek et al. 2022
		<i>N. gonorrhoeae</i>	Azithromycin		Miura et al. 2022
		<i>N. meningitidis</i>	Penicillin		Saito et al. 2022
		<i>C. jejuni</i>	Ampicillin, tetracycline, and ciprofloxacin		Sasaki et al. 2022
Japan		<i>S. Corvallis</i>	Streptomycin		Sasaki et al. 2022
		<i>S. braenderup</i>	Streptomycin		Sasaki et al. 2022
		<i>E. coli serotype O25B</i>	Fluoroquinolone		Matsumoto et al. 2022

molecules that are involved in pathogenesis and deletion of which would impair bacterial viability, making them more vulnerable to the host immune system and susceptible to the prescribed antibiotics. Indeed, it has been suggested that, while less essential for survival, such targets are less likely to cause mutations. Furthermore, it has been suggested that tiny compounds that binds to these "non-essential" genes could be coupled with preexisting antibiotics to broaden therapeutic targets efficacy (Annunziato 2019).

β -lactamase inhibitors

Antibiotics used with β -lactam class (BL) continued to remain the most effective against most of the bacterial strains. Unfortunately, the introduction and modification of β -lactamases offer long-term health risks, which limit the medical efficacy of all currently available β -lactams. The development of inhibitors of β -lactamase is effective in regaining the action of β -lactam medicines can be a good approach (Vrancianu et al. 2020). ABR is a complex phenomenon, with various pathways linked to β -lactam failure including, counting the formation of β -lactamases (that hydrolyze BLAs), structural changes in penicillin binding proteins, lowered the expression of outer membrane porins, and enhanced drug efflux (Drawz and Bonomo 2010). It has been concluded that BLIs can be administered in conjunction with antibiotics to prevent the infections caused by the Gram-negative bacteria (Vrancianu et al. 2020). Further, development of new potent inhibitor formulations and their validation in clinical trials has been suggested.

The discovery of the inhibitor clavulanic acid has led to the cure of several infections caused by β -lactamase-producing pathogens. Clavulanic acid is the inhibitor of most class A β -lactamases and is conjugated with penicillins (amoxicillin-clavulanate) (White et al. 2004). Avibactam is a powerful inhibitor of class A and C β -lactamases that can reversibly revert the serine β -lactamases active site. The synergistic approach of avibactam in combination with ceftazidime has already been accepted for the treatment of abdominal, urinary tract infections and pneumonia/Ventilator Associated Events (VAE) (Tuon et al. 2018; Palwe et al. 2021). Further, it is important to note that M61, M21, M17, M1 inhibitors (Peptidase family contains an aminopeptidase secreted by *Sphingomonas capsulata*) against NDM-1 type metallo- β -lactamases showed promising effect and can serve as potential candidates to enhance efficiency of antibiotics against MDR bacteria (Khan et al. 2017). It has been reported that three novel compounds namely D2573, D2148 and D63 served as a potential non- β -lactam broad-spectrum inhibitors against β -lactamase. These compounds bind with the residues in active site

and do not allow lactamases to hydrolyze the β -lactam antibiotics (Farhat et al. 2022). Various β -lactamase inhibitors with their mode of action which are used to treat infections are listed in Table 2.

Efflux pumps inhibitors

Many organisms developed a strategy to extrude the drug compounds from inside of cell to the external environment through efflux pumps, which confers ABR. Efflux pumps not only are involved in developing resistance through extruding but also help in biofilm formation (Ugwuanyi et al. 2021). In case of prokaryotes, efflux pumps can be classified into five primary families based on their functionality. These families include small MDR family, resistance nodulation cell division (RND) family, ATP binding cassette (ABC), multidrug and toxin extrusion (MATE) family, and major facilitator superfamily (MFS) (Sharma et al. 2019). In this regard, ABC efflux pumps extrude the antibiotics by utilizing the energy derived during hydrolysis of ATP and rest of the efflux pumps extrudes antibiotics by secondary active transport (Blair et al. 2014). The efflux pump inhibitors (EPI) play a crucial role in restoring the activity of various antibiotics by accumulating them inside the resistant bacteria that previously extrudes these antibiotics. It has been reported that *Klebsiella* among the ESKAPE has developed the resistance against most of the drugs (tetracycline, norfloxacin, ciprofloxacin, chloramphenicol) due to the presence of various efflux systems genes like *acrAB* (Acriflavine resistance), *kdeA* (Krypton efflux transporter), *kmrA* (*Klebsiella* MDR), *kpnEF* (*K. pneumonia* efflux pump), and *oqxAB* (Olaquinox and Quinolone efflux pump) (Ashwath and Sannejal 2022). Various reports have also suggested the use of geraniol, reserpine, flavonoids, and catechin gallates (plant derived EPIs) and synthetic efflux inhibitors like CCCP (Carbonyl cyanide-chlorophenylhydrazone) and PA β N (Phenylalanine-arginine β -naphthylamide) in restoring the *K. pneumonia* sensitivity to antibiotics (Ashwath and Sannejal 2022). EPIs like 5'-MHC (5'- methoxyhyd-nocarpin), carnolic acid, carnolol, 4-(2-piperidin-1-ylethoxy)- 2- PPQ (4 propoxyphenyl quinoline) etc play crucial role in reversing the effect of acquired resistance against berberine, erythromycin, tetracycline, norfloxacin, ciprofloxacin in case of *S. aureus* by targeting NorA (Norfloxacin efflux pumpA), MsrA (Macrolide – specific efflux pump A), and TetK (Tetracycline resistance protein K) efflux pumps (Ashwath and Sannejal 2022; Zechini and Versace 2009). The positive role of PA β N to inhibit AdeFGH (Adenine DNA glycosylase Formamido pyrimidine

Table 2. β -Lactamase inhibitors with their mechanism of action.

β -Lactamase Inhibitors	Chemical Structure	Mechanism of action	Reference
Avibactam		A non- β lactam β -lactamase inhibitor that inactivates Ambler class A β -lactamases, class C and some class D β -lactamases by a unique covalent and reversible mechanism	National Center for Biotechnology Information, 2022a
Tazobactam		Irreversible inhibition of beta-lactamase enzymes. Tazobactam may bind covalently to plasmid-mediated and chromosome-mediated beta-lactamase enzymes.	National Center for Biotechnology Information, 2022b
Relebactam		Inhibits Ambler class A, class C enzymes. Relebactam binds non-covalently to the beta-lactamase binding site, then, it covalently acylates the serine residue in the active site of the enzyme.	National Center for Biotechnology Information, 2022c
Sulbactam		It is an irreversible inhibitor of β -lactamase	National Center for Biotechnology Information, 2022d
Clavulanic acid		Clavulanic acid contains a beta-lactam ring in its structure that binds in an irreversible fashion to beta-lactamases, preventing them from inactivating certain beta-lactam antibiotics	National Center for Biotechnology Information, 2022e

Helicase efflux pump) which extrude chloramphenicol, trimethoprim and clindamycin was found in the case of *A. baumannii* (CortezCordova and Kumar [2011](#)). The efflux pump AcrAB-TolC of *E. aerogenes* expel norfloxacin that contributes to its resistance property but this action was prevented by using pyridoquinolines (Chevalier et al. [2001](#)). A wide range of EPIs have been found including natural derivatives and synthetics that work against various efflux pumps which impart bacterial intracellular resistance against large number of antibiotics. The RND superfamily of efflux pumps is commonly expressed in *P. aeruginosa*. It has been observed that MexAB-OprM which belongs to RND superfamily, is responsible for antibiotic resistance including β -lactams, fluoroquinolones, macrolides, aminoglycosides, chloramphenicol, tetracyclines, novobiocin, sulfonamides, tigecycline, and trimethoprim in *P. aeruginosa*. It has been observed that MexAB-OprM is

universally expressed across all strains and plays a key role in the emergence of MDR in case of *P. aeruginosa* (Ahmadian et al. [2023](#)). It was suggested to develop the EPIs particularly to regain the activity of the antibiotics against resistant bacteria (AlMatar et al. [2021](#)). There are several criteria which determine the successful designing of an EPI that includes: EPI should be selective to bacterial efflux pump and not disturb pharmacological activity of eukaryotic pumps, their pharmacokinetic profile and therapeutic index should be good, and it must be feasible economically (Nguyen et al. [2023](#)). Few examples of EPI with their mechanism of action are mentioned in [Table 3](#).

Outer membrane permeabilizers

In case of Gram-negative bacteria, ABR property is naturally present. This is because their outer

Table 3. Examples of EPIs and their mechanism of action.

EPI	Chemical Structure	Mechanism of action	Reference
phenylalanine arginyl β -naphthylamide (PA β N)		Competitive inhibition with antibiotics	Askoura et al. 2011
carbonyl cyanide 3-chlorophenylhydrazine		uncoupler of oxidative phosphorylation	Sanchez-Carbonel et al. 2021; PubChem, 2004a
NSC-33353		Binding to the membrane fusion protein AcrA.	D'Cunha et al. 2021; PubChem, 2004b

membrane comprising lipopolysaccharides (LPS) and phospholipids that plays an important role as permeation barrier. This arrangement of layers prevents entry of hydrophilic drug compounds from the outer surface. In addition, outer membrane proteins (OMP) also help in emitting out hydrophilic compounds. The arrangement of OMPs in Gram-negative bacteria contributes to their ability to develop ABR, making them more challenging to treat with antibiotics as compared to Gram-positive bacteria. In the light of challenge, WHO has identified *A. baumannii*, *P. aeruginosa*, and *Enterobacteriaceae* as the highest-priority bacterial pathogens for the development of new antibiotics. To enhance the efficacy of antibiotics, it is imperative to improve drug diffusion by overcoming the barrier posed by LPS. This problem could be solved by using permeabilizers with property to weaken the outer membrane, so that permeation of antimicrobial agents can be increased. In this way, the influx of drug could be increased which was earlier restrained by the bacterial structure. Permeabilizers help in disintegration of LPS layer leading to disturbance in lipid fraction of outer membrane causing increase in efficacy of antibiotics toward its target. It has been proposed that a synergistic approach of permeabilizers with the antibiotics could provide an unfailing strategy for controlling growth of microbes (Farrag et al. 2019; Klobucar et al. 2021).

Various compounds have shown to increase the outer membrane permeability. For instance, Liprostatin-1 and MAC-0568743 are the compounds which can interact with LPS and help in the disruption of outer membrane. This

helps in potentiating the effect of rifampicin, novobiocin, erythromycin, and linezolid, which were previously found ineffective against Gram-negative bacteria because of their insufficiency to penetrate the cell (Klobucar et al. 2021). Polymyxins are regarded as last-line treatment for the infections caused by Gram-negative bacteria. They not only serve as potent antibiotics but also outer membrane permeabilizers. However, the emergence of strains resistant to these last resort drugs is becoming a critical issue in a growing number of countries (Jeannot et al. 2017). To overcome such problems related to the use of polymyxins against Gram-negative bacteria, combination therapy has been proposed. The combined use of either polymyxin B nanopeptide (PMBN), polymyxin B (PMB), or Ethylenediaminetetraacetic acid (EDTA) with the robenidine analog NCL195 demonstrated synergistic activity against *P. aeruginosa*, *E. coli*, *K. pneumoniae*, and *Enterobacter* spp (Otto et al. 2019; Ramirez et al. 2022, Pi et al. 2020). According to our perspective, synergistic approaches hold great promise for the development of novel drug classes to effectively treat infections caused by Gram-negative bacteria.

Lytic transglycosylase inhibitors (Its)

These play important role in peptidoglycan recycling, metabolism and cell division. Because of their considerable redundancy in Gram-negative bacteria, they have been underexplored as putative antibacterial targets. For instance, O-sulphonated glycopeptide, Bulgecin A predominantly targets soluble lytic transglycosylases (Slt), which was found to enhance the

effectiveness of β -lactam antibiotics that target penicillin binding proteins (PBPs). The LtgA from *N. meningitidis* is a membrane bound homolog of *E. coli* Slt. Bulgecin A is a considerable inhibitor of LtgA. It also additionally restores the efficiency of β -lactam antibiotics in case of *N. meningitidis* and *N. gonorrhoeae* strains which have almost lost sensitivity to β -lactams. Authors have concluded that this inhibitor is particularly relevant to treat *N. gonorrhoeae* infections in humans for which no vaccines are available (Williams et al. 2017). By considering the structural and mechanistic mode of inhibition of Ltgs by Bulgecin A, it has been proposed that Ltgs can be a promising candidate for the development of broad-spectrum inhibitors (Martinez-Bond et al. 2022).

Bacteriophages

Phage therapy involves the utilization of bacteriophages (bacterial viruses) to combat bacterial infections. Phage therapy has been in practice for nearly a century. Bacteriophages are highly selective to species or strains, preventing dysbacteriosis and secondary infections. Where target bacteria are present, phages self-amplify, enhancing the local antibacterial actions—classical predator-prey mediated ecological equilibrium. Unlike traditional antibiotics, phages are able to invade and eliminate biofilms (Kortright et al. 2019). Therapeutics which were available to treat infections are no longer effective because of the rise in AMR. Observing this situation, European medical and scientific societies are looking for options in the form of phage therapies to circumvent the problem of AMR. Bacteriophage therapy provides several benefits, which are to be used as an alternate therapy for combating resistance. For instance, synergistic approach of phage therapy with antimicrobials provides an efficient approach toward MDR pathogens. Phage therapy is narrow spectrum hence has more sensitivity toward pathogenic bacteria and also have no impact on bystander flora (De Sordi et al. 2017). Lytic bacteriophages have capability to replicate once they get to the infectious site, hence they accumulate and selectively eradicate the bacterial population (Nikolich and Filippov 2020). Phage display has been suggested as useful, economical, and fast technique in order to identify immunogenic proteins and peptides which was found to be useful for the development of novel and more effective vaccines. Biofilm formation which is the mark of severe infections could be disintegrated by various enzymes inherited by these phages, allowing the antibiotics to cross the biofilms and cause the obstruction of crowded pathogens. In our views, phage

therapy could serve as an economically viable alternate against MDR pathogens as compared to stand-alone antibiotics (Merril et al. 1996). On a head-to-head trial in broiler chickens, the efficiency of standard antibiotics and phage therapy as a single treatment and in combination of the two has been examined (Huff et al. 2004). It has been observed that upon using as a single treatment, antibiotic Enrofloxacin (a fluoroquinolone) was found to decrease the mortality rate from 68% in untreated birds to 3% in treated birds. In comparison, phage treatment as a standalone intervention resulted in a mortality reduction to 15%. It is important to note that there was no mortality found when phage and enrofloxacin therapy was used in combination (Huff et al. 2004). In case of *P. aeruginosa* which causes endocarditis in rats, it has been observed that the combination of phage therapy and antibiotic ciprofloxacin treatment reduced bacterial burden many folds than phage or ciprofloxacin treatment alone (Oechslin et al. 2017). It has been suggested that bacteriophages can be effective as a therapy against intracellular pathogens. It has been observed that in the case of intracellular pathogens such as *Mycobacterium tuberculosis*, bacteriophage therapy can be a promising strategy as an alternate or complementary treatment of antibiotics. It has been concluded that bacteriophage microencapsulation and artificial vectorization are good for efficient and targeted intracellular delivery to kill intracellular pathogens, therefore can be used as an alternate to antibiotics (Nieth et al. 2015). As phages can move through transcytosis of epithelial cell layers and can spread throughout sterile regions of human body organs including the brain, hence in our opinion they can be used for human health and immunity as an alternate to antibiotics (Nguyen et al. 2017).

Nanoparticles to combat the microbial resistance

There exist several nanoparticles mediated mechanisms to fight against the microbial resistance including nanoparticle releasing nitric oxide (NO NPs), nanoparticles containing chitosan (Chitosan NPs) and different metals. In this relation, NO NPs generate reactive nitrogen oxide intermediates (RNOS) which reacts with plasma membrane proteins of bacteria, inactivates zinc metallo-proteins, causes lipid peroxidation, which results in damage of DNA molecules either by deamination, production of hydrogen peroxide or inhibiting DNA repair enzymes (Pelgrift and Friedman 2013). Chitosan NPs are derived from chitin which is positively charged polysaccharide because of which they can bind to the negatively charged cell wall of bacteria causing increase in cell wall permeability leading to

osmotic damage. Chitosan also inhibits protein translation by inhibition of mRNA transcription after binding to bacterial DNA (Pelgrift and Friedman 2013). It has been found that metal nanoparticles have vast inhibitory properties against bacteria, viruses and fungi as they generate damage to microbial cell membrane, protein and DNA and found to generate ROS species causing oxidative stress (Figure 2). They could also be used as efficient drug carriers. Additionally, metal NPs like silver, gold, copper oxide, zinc oxide, magnesium oxide and titanium oxide particles can be used as antimicrobials agents (Brandelli et al. 2017). Inorganic nanomaterials can also serve as antimicrobials and also as antibiotic delivery system. Various types of inorganic nanoparticles are considered significant, including metal nanoparticles (NPs), carbon-based nanoparticles (such as carbon nanofibers and carbon nanotubes), semiconducting materials (such as CdSe, CdS, ZnS, TiO₂, PbS, InP, Si/SiO₂), magnetic materials (such as Fe₃O₄, Co, CoFe₂O₄, FePt, CoPt, and their composites), as well as lanthanide materials (such as Gd₂O₃ and Eu₂O₃) (Sharma et al. 2012). In future, these nanoparticles can serve as a therapeutic agent to kill bacterial populations. For instance, it has been suggested that Zinc Oxide nanoparticles (ZnO-NP), polyvinylpyrrolidone (PVP)-capped AgNPs, gallate-polyvinylpyrrolidone-capped hybrid silver nanoparticles (G-PVP–AgNPs) can be used as an alternate therapy to antibiotics in order to control the infections caused by *A. baumannii* which has developed resistance particularly against

carbapenems (beta-lactam antibiotics). In our views these three studies are very important as *A. baumannii* is an opportunistic pathogen with multi-drug resistance capabilities, posing a significant threat and potentially resulting in increased mortality rates (Tiwari et al. 2017; 2018; 2020).

Targeting of host pathway to combat infections

Host-directed therapy (HDT) is a nascent strategy aimed at combating AMR (Kaufmann et al. 2018). It involves targeting host cell factors essential for pathogen survival, particularly in terms of replication and persistence. Furthermore, HDT encompasses the manipulation of host cells to enhance immune responses against pathogens, mitigate excessive inflammation, and restore immune reactivity balance at infection sites. For instance, infection caused by Group A Streptococcus (GAS), which gives rise to a spectrum of severe diseases, including acute rheumatic fever (ARF) and rheumatic heart disease (RHD) primarily affecting children aged 5-15 years. Each year, GAS is responsible for 616 million cases of pharyngitis, 18.1 million severe cases, and 517,000 deaths worldwide (Carapetis et al. 2005). However, the development of vaccines is impeded by the substantial number of serotypes associated with GAS (Sanduja et al. 2020). As a countermeasure, antibiotics such as penicillin and cephalosporins, among others, are utilized to combat various GAS diseases. Nevertheless, the emergence of

Figure 2. Schematic representation depicting role of nanoparticles (NPs) to combat microbial resistance. NO NPs generate RNOS, damaging bacteria proteins and DNA. Chitosan NPs, derived from chitin, bind to bacteria cell wall, increasing permeability and inhibiting protein translation. Metal nanoparticles damage microbial membranes, proteins, and DNA, acting as potent antimicrobials and drug carriers.

ABR in certain GAS clinical isolates against macrolides and tetracyclines across different regions has become a global concern (Cattoir 2016). The successful establishment of an infection relies on the pathogen's ability to acquire essential nutrients from the host. In case of GAS, it has been observed that during adherence to host cells, GAS releases streptolysin toxins, leading to endoplasmic reticulum stress. Consequently, this stress induces increased expression of asparagine (ASN) synthetase and subsequent production of ASN. The bacteria detect the released ASN, which triggers alterations in the expression of approximately 17% of GAS genes. Notably, the expression of streptolysin toxins was found to be significantly upregulated, while genes associated with proliferation are downregulated in the absence of ASN. Interestingly, asparaginase, a commonly used chemotherapeutic agent, effectively halts GAS growth in human blood and inhibits GAS proliferation in a mouse model of human bacteremia. These findings unveil a pathogenic pathway and propose a novel therapeutic strategy against GAS infections. Furthermore, it has been established that GAS-mediated asparagine induction and release occur *via* the PERK-eIF2 α -ATF4 branch of the unfolded protein response. Treatment of infected mice with ISRIB or GSK2656157 inhibitors has demonstrated the termination of GAS infections (Anand et al. 2021) (Figure 3). Therefore, we suggest the use of these inhibitors as an alternate of antibiotics for the treatment of GAS infections, however it needs further clinical trials.

It is noteworthy that in 2015, MDR tuberculosis (TB) and extensively drug-resistant (XDR) TB accounted for over 10% of TB-related fatalities (WHO 2023b). This emergence of drug-resistant TB can be attributed to suboptimal patient adherence to a six-month chemotherapy regimen involving drugs such as isoniazid, rifampin, ethambutol, and pyrazinamide. Moreover, the genetic diversity and clonality of *Mycobacterium tuberculosis* (Mtb) within patients, (Eldholm and Balloux 2016) along with reduced drug penetration into lesions, can lead to ineffective monotherapy at sites where bacterial colonization occurs, despite treatment with multiple drugs (Moreno-Gamez et al. 2015; Lenaerts et al. 2015). These factors contribute to the emergence of AMR in the case of tuberculosis. Of equal concern is the rapid acquisition of resistance to newly approved drugs for MDR TB, such as delamanid and bedaquiline (Bloemberg et al. 2015). In our view, given the development of AMR and the limited availability of drugs for MDR TB and XDR TB, there is a strong need for host-directed therapy as a primary adjunct to conventional chemotherapy.

Moreover, in case of sepsis, numerous pathways that impair the host's ability to defend against the condition

are activated. The expression of programmed cell death 1 (PD1) is increased in circulating T cells, and animal models of sepsis have demonstrated that administration of PD1-blocking antibodies, as well as PD1 ligand (PDL1)-blocking antibodies, improves survival (Kaufmann et al. 2018). In addition, several pathogenic bacteria and toxins exploit host cytosol access by trafficking through acidified endosomes, facilitating translocation through the host membrane. Blocking host receptors and entry pathways utilized by toxins represents another promising therapeutic approach, as significant adaptations would be required by the microbe to switch to a new receptor capable of supporting pathogenesis while evading intervention. Toxins such as *P. aeruginosa* ExoA and diphtheria toxin (DT) exert their effects on cells by ADP-ribosylation of EF-2, thereby interrupting the translation process. A compound known as 4-bromobenzaldehyde *N*-(2,6-dimethylphenyl) semicarbazone (EGA) has been reported to impede the translocation process of acid-dependent bacterial toxins in mammalian cells. Likewise, ADAM 10, a cell surface metalloprotease, serves as a receptor for the alpha-toxin produced by *S. aureus*. The inhibition of ADAM 10 protects against the pathogenic effects induced by alpha-toxin in murine staphylococcal pneumonia (Gillespie et al. 2013; Munguia and Nizet 2017). Additionally, mitoxantrone (MTX) can function as host-targeted immunotherapy for wound infections caused by vancomycin-resistant *E. faecalis* (VRE). This therapy involves the clearance of enterococcal bacteria by macrophages. In a mouse model of wound infection, MTX treatment effectively recruits and activates macrophages, leading to bacterial clearance (da Silva et al. 2023).

CRISPR-cas9 as a next generation tool

In our opinion, CRISPR-cas system could act as a next generation tool to cope up with the problem of ABR, especially treating superbugs. The CRISPR-Cas9 serves as a DNA cutter which is guided by the RNA, it's a tool which is based on the bacteria's immune system to deal with the foreign particle like bacteriophages or mobile genetic elements. Upon invasion of foreign genetic material inside host cell, the CRISPR system adds the foreign gene in bacteria's genome known as spacer whose transcription directs Cas protein to cleave and secure bacteria from bacteriophage or any other foreign complementary genetic material. The CRISPR-Cas has an advantage that it can differentiate between pathogenic and commensal strains as it is highly sequence-specific. We can design the guide CRISPR-RNA against virulence gene or the resistance gene in pathogenic bacteria therefore using bacteria's own defending tool to use against the bacteria itself, resulting in dsDNA break in the gene and converting

Figure 3. (A) GAS adheres to the mammalian cell membrane and releases streptolysin O (SLO) and streptolysin S (SLS) in cells. These toxins induce ER (endoplasmic reticulum) stress and activates the unfolded protein response (UPR). An ER transmembrane protein: protein kinase RNA (PKR)-like ER kinase (PERK) gets phosphorylated in presence of activated UPR due to SLO and SLS. PERK phosphorylates α subunit of the eukaryotic translation initiation factor (eIF2 α) at Ser51 causing inhibition of global translation and activates ATF4 (activating transcription factor 4). ATF4 controls the expression of asparagine synthetase and thus increases the production of asparagine. The asparagine is transported out of the cell and the extracellular asparagine is sensed by TrxSR of GAS. Asparagine causes the alteration of transcription of 17% GAS genes and thus increases its multiplication rate. (B) GSK26 and ISRIB both inhibits the GAS proliferation. GSK26 is a selective inhibitor of PERK; ISRIB stimulates global translation in presence of phosphorylated eIF2 α by activating eIF2B.

resistant bacteria to sensitive. However, designing guide RNA is a careful process as it can bind to nonspecific target and may impart dysbiosis. Hence designing should be done in such a way that it avoids maximum off-target

effects (Aslam et al. 2020). Recently, a conjugative plasmid, pPD1, with a CRISPR-Cas9 expressing cassette was effectively transferred to *E. faecalis* for the selective removal of two genes namely, erythromycin resistance (*ermB*) and

tetracycline resistance (*tetM*) which impart resistance to erythromycin and tetracycline respectively. The transformation of the recombinant pPD1, with a CRISPR-Cas9 resulted in the reduction of resistant *E. faecalis* in the murein intestine (Rodrigues et al. 2019).

Delivery methods for CRISPR–Cas9 system

There exist several delivery strategies for CRISPR-Cas9 system to combat ABR problem such as (i) non-viral vector-based delivery (ii) phage-based delivery (iii) nanoparticle-based delivery. For CRISPR-Cas9 delivery, lipid nanoparticles are most relevant and most extensively chosen delivery vehicle some of which are also clinically approved for trial as they not only encapsulate the CRISPR DNA, guide mRNA but also protect the RNA from RNases during penetration of cell membrane (Wan et al. 2021). Delivering the gene to the specific site is always a crucial step for any delivery systems, to overcome such a problem, a strategy has been developed known as selective organ targeting (SORT) and delivered lipid NPs containing Cas9 mRNA/sgRNA (single guided RNA) to liver, lung and spleen (Cheng et al. 2020). Current studies also demonstrated that temperate phage could also be used to deliver plasmid or phagemid containing a CRISPR/Cas9 system directed to target an ABR gene or a specific pathogen (Park et al. 2017). Further it was concluded that this strategy could selectively control ABR strains of *E. coli* or MRSA with minimal effects on naïve bacteria. Additionally, it has been suggested that non-viral vector methods like electroporation, microinjection, cell-penetrating peptide could also be used to deliver CRISPR–Cas9. In this regard, electroporation is suitable for plasmid-based approach where either mixture of Cas9 mRNA and sgRNA or Cas9/sgRNA RNPs could be delivered by increasing the permeability of the cell. Microinjection of Cas9 mRNA and sgRNA is another approach of delivery. In this case, foreign material can be directly injected into living cells (Figure 4). On the other hand, cell penetrating peptides (CPP) method can also be used to translocate across cell membrane. In this regard, Cas9 protein can be conjugated with CPP molecule and sgRNA via electrostatic interaction and allowed to transfer to cell (Liu et al. 2017).

Plant derivatives

Medicinal herbs are always considered as sleeping giants, as their chemical properties make them a good source of natural drugs. Medicinal herb could help in solving the global problem of AMR and could deal

with the menace created by medicines in market. Only fluoroquinolones, oxazolidinones, and sulfonamides classes of antibiotics are considered as completely originated through synthetic chemistry while six out of total nine classes have compounds which are naturally originated. It's very necessary to search for plant based medicinally extracted compounds. Plant based medicinal compounds can either be used alone or in combination with antibiotics to cope up with AMR imparted crisis. It has been found that extracted of *Crocus sativus* commonly known as saffron has antimicrobial and antifungal activity (Butnariu et al. 2022). According to this study modified compounds of *C. sativus* have promising antimicrobial activity (Carradori et al. 2016). Plant product like berberine has been shown to serve as an inhibitor of efflux pumps (Lewis and Ausubel 2006). Moreover, the utilization of berberine derived from the *Argemone mexicana* L. plant has exhibited inhibitory effects on the growth of various bacteria, including *S. aureus*, *B. cereus*, *E. coli*, and *P. aeruginosa* (More et al. 2017). This finding suggests that ethnopharmacology can guide the exploration for effective medicinal compounds to mitigate the proliferation of ABR. Given the global impact of this issue on health and economy, it is imperative to urgently seek new alternatives and formulate novel policies to combat ABR (AISheikh et al. 2020). Consequently, the development and application of plant-based antimicrobials emerge as a crucial necessity to counteract ABR and restore sensitivity in resistant microorganisms. These studies shows that structurally or chemically modified compounds can have more efficient, effective and promising results and can help in the development of some novel drugs in combating with an antimicrobial situation.

Combination therapy

Considering the current situation, antibiotics combination therapies may provide potential therapeutic possibilities to control infections, providing a proper experimental standardization of such formulations with *in vitro* and *in vivo* validation. However, combination of antibiotics should meet criteria of synergistic response, capability to restrain the emergence of spontaneous resistance and reduced antibiotic toxicity to the host cells (Liu et al. 2021). Combination of antibiotics usually recommended to treat critically ill patients suffering with septic shock and severe sepsis caused by the MDR pathogens. Therefore, based on current situation of a faltering antibiotic production pipeline, the approach of combination therapy may be used, and its potential can be further explored to enhance the

Figure 4. Different delivery methods for CRISPR–Cas9 system. Nanoparticle–based delivery: Cas9–gRNA ribonucleoprotein (RNP) can be transferred in cell with the help of nanoparticles. Plasmid–based delivery: Cas9/gRNA plasmid can be transferred through electroporation or microinjection. Cas9 mRNA is translated into Cas9 protein which forms RNP complex with gRNA which in turn edits the targeted resistance gene. Phage–based delivery: Phage particles are used to deliver the coding sequences which code for CRISPR–Cas9 system.

antibacterial effects of available antibiotics. There are several studies proposing combination antibiotic therapies against MDR bacteria. Earlier studies have been performed on combinations of antibiotic of different classes for example, combination of streptomycin with cefotaxime, doripenem with cefotaxime, and aztreonam with cefotaxime can be used against *bla*_{CTX-M-15} β -lactamase producing bacterial strains (Maryam and Khan 2016; Maryam and Khan 2017; Maryam and Khan 2018). Whereas combination of antibiotics which can be effective against *bla*_{NDM-1} producing bacteria, are reported as doripenem with cefoxitin as well as, doripenem with tetracycline (Maryam et al. 2019). The combinations of streptomycin with amikacin and streptomycin with ciprofloxacin have also been proved effective against *bla*_{NDM-1} producing bacteria (Maryam et al. 2020). Other potential combinations against MDR bacteria may also be screened in future to increase therapeutic options. Moreover, various studies have also shown that a combination of cell penetrating–peptides with antibiotic, can be effective against several resistant pathogens (Maryam et al. 2021). Bacteria have multiple resistance options, like antibiotic target modification, over–expression of efflux pump and

inactivation of antibiotics, may be treated by giving combination therapy (Figure 5). In our opinion, other potential combinations against MDR bacteria may also be screened in future to increase therapeutic options.

New therapeutic agents and clinical trials

In the realm of antibiotic development, significant progress has been made across various phases of clinical trials. It is important to note that nafithromycin which is a next generation lactone ketolide antibiotic slated to enter phase III clinical development in India for the treatment of community acquired bacterial pneumonia. This antibiotic showed promising activity against MDR *S. pneumoniae* including erythromycin and telithromycin-resistant resistant strains (Chavan et al. 2021). A phase III clinical trial (SURE-1) was performed in women patients infected with fluoroquinolone-susceptible pathogens showed, that sulopenem was not non-inferior to ciprofloxacin. Further, a phase III clinical trial (SURE-2) compared intravenous sulopenem followed by oral sulopenem etzadroxil/probenecid with ertapenem in the treatment of complicated UTIs. In this trial no difference in overall

Figure 5. Diagrammatic description of antibiotic resistance in monotherapy vs combination therapy against antimicrobial resistance wherein after use of adjuvant, there can be enhancement of activity of antibiotic. Adjuvant whether it is β -lactamase inhibitors, efflux pump inhibitor or cell membrane permeabilizer all can help in regaining the efficacy of resistant antibiotics.

success was observed at the end of therapy. FDA, Iterum stated that they are currently evaluating the optimal design for an additional phase III uncomplicated UTI study to be conducted prior to the potential resubmission of the New Drug Application (NDA). The safety and tolerability of sulopenem has been reported in various phase I pharmacokinetic studies and phase III clinical trials. It is important and crucial to note that Sulopenem (intravenous and oral) found to be well tolerated in healthy patients (Zhanel et al. 2022).

Moreover, promising evidence has surfaced for the effectiveness of taniborbactam (VNRX-5133) in conjunction with cefepime against carbapenem-resistant *Pseudomonas aeruginosa* (CRPA) (Liu et al. 2020). Notably, positive top-line outcomes have been disclosed for durlobactam (ETX2514) paired with sulbactam in a Phase III trial for treating carbapenem-resistant *A. baumannii* (CRAB) infections (McLeod et al. 2020). Additionally, there have been notable advancements with seven new products entering in Phase 1 trials. Amongst, XNW4107, studied alongside imipenem and cilastatin, is positioned as a combination therapy encompassing a β -lactamase inhibitor (BLI), exhibiting potential activity against CRAB, CRPA, and carbapenem-resistant *Enterobacterales* (CRE) (Clinical Drug Experience Knowledgebase 2023). QPX7728+QPX2015, a boronate BLI combination likely with a β -lactam, has entered Phase 1, aiming to combat CRE (Lomovskaya et al. 2021). Further, two new

polymyxins-MRX-8 and QPX9003—have entered Phase 1 trials focusing on treating Gram-negative infections (Lepak et al. 2020; Aslan et al. 2022). An undisclosed compound, RG6006 (Abx MCP, macrocyclic peptide), is also entered in Phase 1 trials targeting *A. baumannii* (Walesch et al. 2023). Additionally, a novel topoisomerase inhibitor, BWC0977, has entered Phase 1 trials with a broad spectrum of therapeutic agents against both Gram-positive and Gram-negative bacteria (ClinicalTrials 2023). Collectively, these new developments have increased the count of traditional agents in Phase 1 trials to a total of 16 (WHO 2022). More clinical data are required to fully assess the clinical efficacy and safety of these new therapeutic agents, especially in patients with complicated UTIs caused by resistant pathogens such as ESBL-producing, Amp-C, MDR *E. coli*. It has been suggested that antimicrobial stewardship programs will need to create guidelines for such trials and treatments (Zhanel et al. 2022; WHO 2022). Table 4 summarizes new antibacterial agents which are under clinical development against WHO priority pathogens.

Conclusion

It is evident from the literature presented that ABR, MDR are global concern leading to the development of untreatable diseases. WHO reported that ABR is among top ten worldwide issues faced by the mankind. Many

Table 4. New antibacterial agents under clinical trials against WHO priority pathogens.

Drug	Anticipated activity against	Antibacterial class	Phase Trial	Reference
Nafththromycin	MDR <i>Streptococcus pneumoniae</i>	Macrolide	Phase 3	Chavan et al. 2021
sulopenem-etzadroxil/probenecid	MDR <i>Enterobacterales</i>	β -Lactam (penem)	Phase 3	Zhanel et al. 2022
Taniborbactam + cefepime	carbapenem-resistant <i>Pseudomonas aeruginosa</i> (CRPA)	Boronate BLI + β -lactam (cephalosporin)	Phase 3	Liu et al. 2020
Durlobactam + sulbactam	carbapenem-resistant <i>Acinetobacter baumannii</i> (CRAB)	Diazabicyclooctane (DBO)-BLI/PBP2 binder + β -lactam-BLI/ PBP1,3 binder	Phase 3	McLeod et al. 2020
XNW4107 + imipenem + cilastatin	CRAB, CRPA, and carbapenem-resistant <i>Enterobacterales</i> (CRE)	BLI + β -lactam (carbapenem) / degradation inhibitor	Phase 1	https://www.cdek.liu.edu/trial/NCT04802863/
QPX7728 + QPX2015	CRE	Boronate-BLI + undisclosed oral β -lactam	Phase 1	Lomovskaya et al. 2021
MRX-8	<i>A. baumannii</i> , <i>P. aeruginosa</i> and <i>E. coli</i>	Polymyxin	Phase 1	Lepak et al. 2020
QPX9003	<i>P. aeruginosa</i> and <i>A. baumannii</i>	Polymyxin	Phase 1	Aslan et al. 2022
RG6006	<i>A. baumannii</i>	Macrocytic peptide	Phase 1	Walesch et al. 2023
BWC0977	Gram-positive and Gram-negative bacteria	Topoisomerase	Phase 1	https://clinicaltrials.gov/study/NCT05088421 .

of the serious infections caused by microbes would remain untreated if no new target or alternatives for antibiotic are identified or developed. To overcome this vast problem cutting edge research is required with some promising outcome. This emerging concern led researchers to identify and develop alternative therapeutic approaches, including inhibitors of efflux pump, β -lactamase and lytic transglycosylases. In addition, permeabilizers for outer membrane can also help in this process. Further, use of bacteriophages and nanoparticles are yet another promising approaches to deal with ABR. In addition, development of inhibitors against host pathways which bacteria uses to get nutrition for proliferation can also be effective approach. Most recently, CRISPR-Cas9 is also considered as an accurate and effective tool to selectively destruct the resistance trait in pathogenic bacteria. Moreover, combination therapy using available antibiotics is quickest, more effective and safe method to combat MDR infections. Sleeping giants of pharmaceutical industries, the medicinal compounds extracted from the medicinal herbs could also assist dealing ABR by opening an avenue to develop novel natural drugs. All these alternative strategies could help resolve ABR. Moreover, this field demands for continuous research to develop novel therapeutic approaches to overcome AMR/ABR/MDR. It is very crucial to mention that more clinical data and antimicrobial stewardship programs are required to fully assess the clinical efficacy and safety of new therapeutic agents. In this regard, Indian Council of Medical Research (ICMR) has issued guidelines to all hospitals, clinics and various other health systems in India to report AMR/ABR/MDR microbes directly to them so that these can be tracked on time (Indian Council of Medical Research 2022, https://main.icmr.nic.in/sites/default/files/upload_documents/AMRSN_Annual_Report_2022.pdf). Moreover, health

care system/infrastructure development and surveillance are very crucial to combat ABR and MDR. Hence, recently during the COVID-19 pandemic, more than 15 new medical colleges in different under-served districts of Uttar Pradesh (UP) state, India under the leadership of Chief Minister of UP and Prime Minister of India, for the surveillance/epidemiology and treatment work have been established. This type of development of infra-structure would be very useful as a model system in order to develop health related facilities to track and tackle the AMR/ABR/MDR. In addition, IMF, WHO and G-20 group can launch more program like GRAM and GLASS in collaboration with the developing countries/economically less developed countries for the surveillance/treatment and to provide more data related to ABR/AMR and MDR.

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